Power BI Assignment 1 – Data Transformation & Data Modeling

Import Data:

Importing data through **GET DATA -> TEXT/CSV.** Before Load, I have import Power Query Editor by clicking “Transform”.

Data Transformation:

● *Restrict the "List of Orders" table to only the first 500 rows.*

Go to Home-> Keep Rows-> Keep Top Rows-> 500.

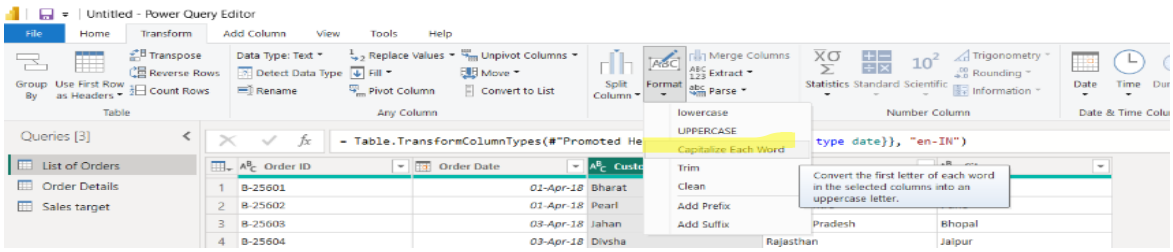
● *Ensure the “Order Date” column in the “List of Orders” table is set to data type 'Date'.*

Right click on the Order Date column-> Choose Type-> Date to change the date format.

● *Change the data type of “Amount” and “Target” columns to ‘Fixed Decimal Number’.*

Press Ctrl and select **Amount** and **Target** columns-> **Right click on Column-> Choose Type-> Fixed Decimal Number** to change the format.

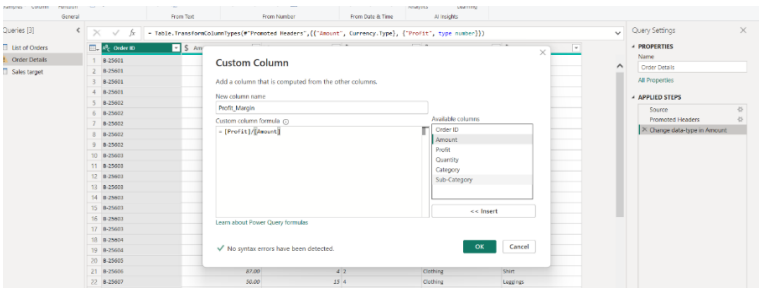
● *Format the **"CustomerName"** column into proper case, ensuring consistent capitalization for each word.*



● *Merge the "State" and "City" columns to create a new column named "Location" in the format ‘City, State’.*

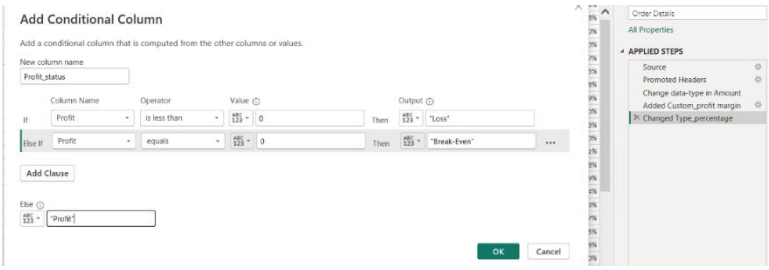
Press Ctrl and select **State** and **City** columns Go to **Add Column-> Merge Column in From Text Group-> Choose Separator and give New Column name.**

● ***Create a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount".***



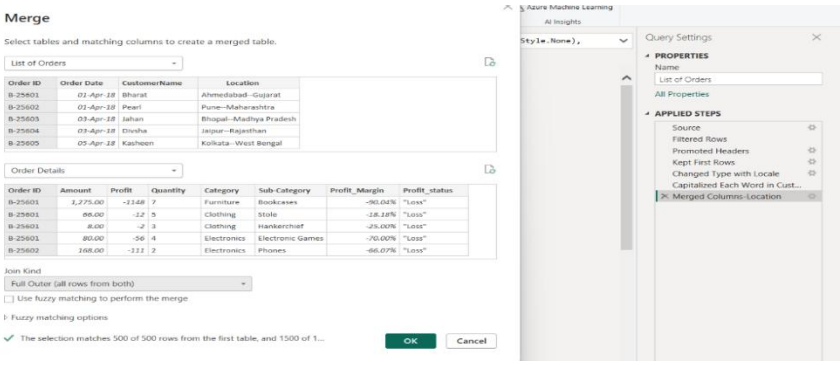
Go to Add Column-> Select Custom Column in General group-> Give New Column Name “Profit_Margin”-> Select Profit and Amount in Available columns pane for doing custom column Formula **(=[profit]/[Amount])** and Click OK.

● *Add a new conditional column named "Profit Status" based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".*



Merging Data (Joins):

● *Merge the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.*

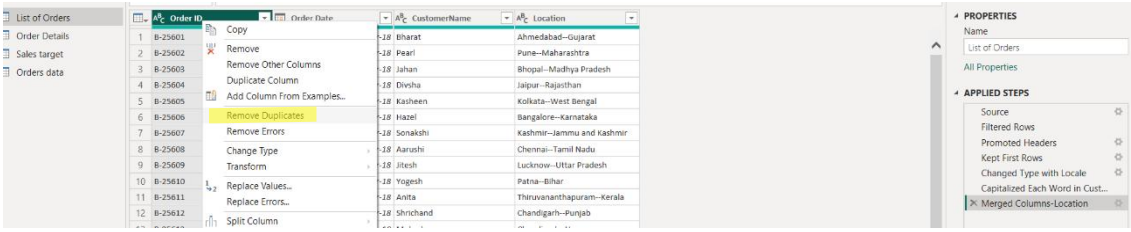


Handling Missing Data & Duplicate Data:

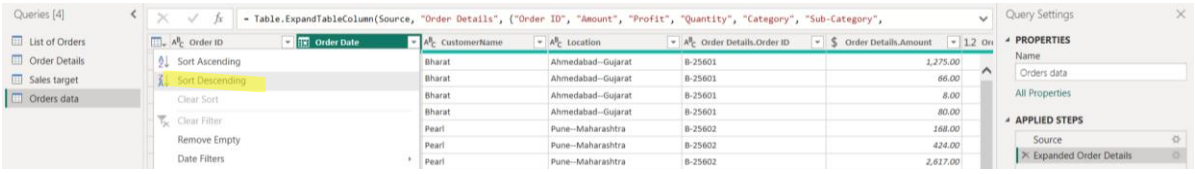
● *Identify missing values in the data and determine a strategy to address them.*

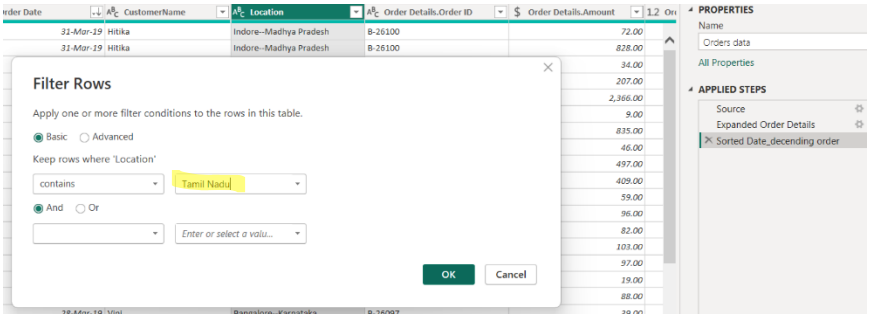
Click Down arrow on top of the column like Order_ID, Sales_ID or else combination of Name, ID, Products, for check is there any NULL or Repeated data. If there is any null in numeric column, take **AVERAGE** that column and fill data in those missing rows. In case, that is Categorical column, take **MODE** and fill.

● *Check for duplicate rows and define a strategy to handle duplicates.*

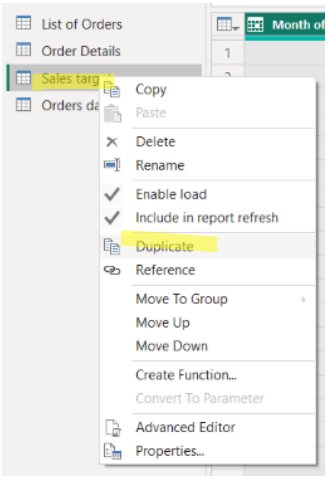


Sorting and Filtering Data:

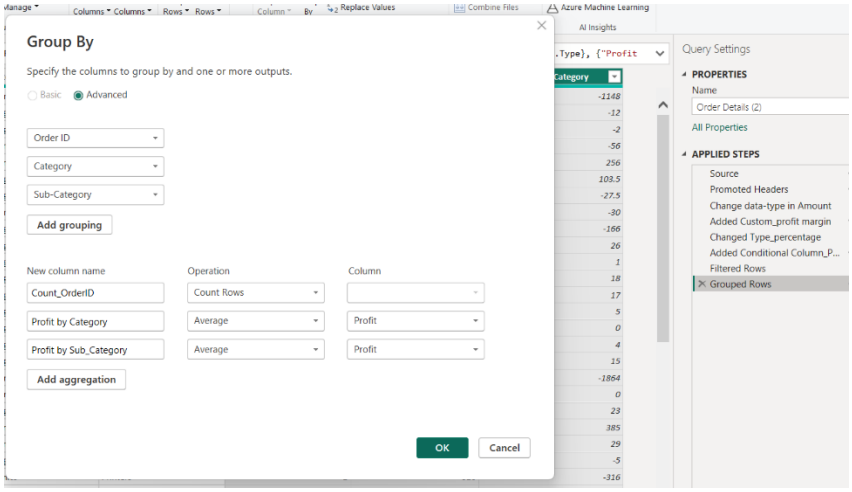




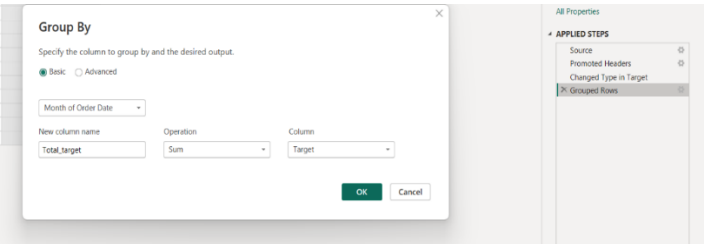
Grouping and Aggregating Data:



Duplicate the "Order Details" table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

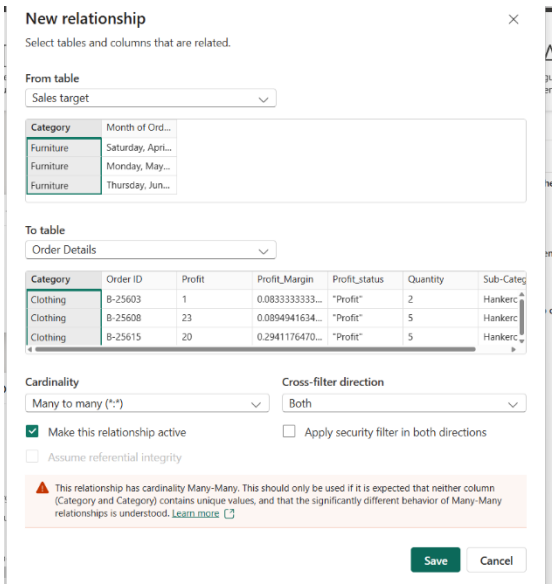
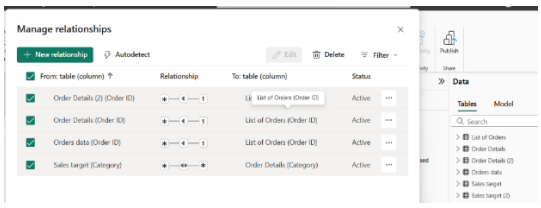


Duplicate the "Sales Target" table and aggregate the total target amount by Month of Order Date.



Data Modeling:

Establish a relationship between the "List of Orders" and "Order Details" tables using the 'Order ID' column.



Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.